

Research paper

Auxetic structure optimization for double-walled hull crashworthiness using response surface methodology

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ABSTRACT

One type of negative Poisson's ratio (NPR) structure, called auxetic, has been widely studied due to their superior mechanical properties, including enhanced energy absorption capabilities. However, their application in marine structures, particularly in improving ship hull crashworthiness, remains underexplored. This study investigates the optimization of an auxetic structure as a filler material in the void space of a double-walled ship hull using Response Surface Methodology. The optimization process considers geometric parameters and material selection. The drop tests was conducted numerically to evaluate energy absorption performance. The results indicate that the optimized model exhibited the highest energy absorption of 2049.91 J, representing an improvement of over 2066 % compared to the least efficient configuration (94.60 J), significantly outperforming other configurations. The findings provide valuable insights into the design of auxetic fillers for maritime safety applications, enhancing the structural integrity of double-walled ship hulls against high-velocity impacts.

1. Introduction

The auxetic structure has negative Poisson's Ratio (NPR) property that defines the tendency of a structure to deform in inward direction perpendicular to the applied force. It means the structure behavior will enlarge under stretching loads, and contract under compression loads [1]. The shape of auxetic structure like re-entrant and double arrowhead had Poisson's Ratio (PR) value in -1.816 and -0.466 [2] that give superior mechanical properties [3] such as high shear modulus [4,5], high indentation resistance [4], improved buckling strength [6,7], high young modulus [8], and high energy absorption property [9,10]. According to Scarpa et al. [11], auxetic structure has an energy-damping capacity that can be up to ten times greater than non-auxetic structure under periodic loading. Xinqiang Lan et al. [12] tested the uniaxial quasi-static compression in the double-arrowhead honeycomb (DAH) auxetic structure which made by a selective laser melting process from Ti-6Al-4 V material with varied relative densities. The maximal specific energy absorption would be achieved by the specimen with high relative densities. Buyun Su et al. [13] investigated the re-entrant honeycombs (RH) auxetic cellular materials with different angles and geometric parameters had significant effect on the anisotropic properties and Poisson's ratio. Jin Zhou et al. [14] compared the quasi-static loading

conditions of hexagonal, re-entrant and double arrowhead auxetic structures. The result showed that the double arrowhead (auxetic) structure has greater specific energy absorbed, such as 125 % and 244 % greater than that of the hexagonal honeycomb (non-auxetic) and re-entrant (auxetic) structures, respectively. Double arrowhead auxetic (DAAs) enhanced energy absorption ability under impacts of high velocity [15,16]. and transverse tensile forces [17]. Avinash Mohan et al. [18]. observed the auxetic structures that were made from plastic polylactic acid (PLA) using Fused Deposition Modeling (FDM) showed that the auxetic structure had a higher compressive plateau stress and energy absorption capacity. This work is notably different from other studies on auxetic structures by focusing specifically on auxetic structures integration into the void spaces of double-walled hull structures. While previous research has largely explored auxetic structures in automotive, aerospace, biomedical, and construction applications [19].

In high-speed marine collisions, double-walled hull structures offer enhanced protection due to their dual-layer configuration, which creates a void space between the outer and inner hulls. While a double-wall hull improves crashworthiness by delaying the penetration of impact forces, the empty space alone does not significantly absorb the energy from a collision [20]. The problem causes substantial deformation, leading to inner hull and compromising the vessel's integrity, particularly if the

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inner hull fails to withstand the transmitted force [21,22]. Xiong et al. developed a double-hull construction capable of preventing damage to the inner wall [23]. Internal girders in horizontal or diagonal reinforcements had been developed within the void space [22–24]. In addition, the influence of void width has been found to reduce the risk of penetration into the inner hull and increasing the outer hull plate thickness can enhance structural strength and delay impact-induced penetration [25–27]. The addition of reinforcement ribs within the hull structure has also proven to improve impact resistance [28]. Despite these efforts, a research gap remains in the utilization of the narrow void space between double hull walls to improve crashworthiness. Compared to an unfilled double-wall hull, the presence of an optimized auxetic filler structure efficiently absorbs and distributes impact energy, reducing localized stress concentrations and minimizing damage propagation [29]. However, integrating an auxetic as filler structure in void space significantly enhances impact resistance.

Therefore, a new approach is needed that not only considers structural strength but also optimizes the geometric configuration of filler materials within the void space between hull walls to maximize energy absorption during collisions. To date, maritime applications have seen limited implementation of Response Surface Methodology (RSM)-based optimization, particularly in the context of structural crashworthiness. While RSM has been widely adopted in fields such as ergonomic designs [30], experimental optimization across various domains [31], prediction the wind speed [32], design optimization [33,34], automotive optimization [35], and manufacturing process [36], its use in ship structural design especially involving advanced meta-materials like auxetic structures remains scarce. Existing maritime studies predominantly focus on conventional reinforcements, material selection, or hull geometry adjustments, often relying on deterministic methods rather than probabilistic or statistical optimization frameworks like RSM. Thus, applying RSM to optimize auxetic geometries within the void space of double-hull structures represents a novel and underexplored area in marine crashworthiness research.

Based on the unique deformation characteristics of auxetic materials, the study hypothesizes that their integration into the void space of a double-hull ship structure enhances crashworthiness during side impacts. Specifically, it is hypothesized that (1) increasing the auxetic cell density and wall thickness improves energy absorption, (2) different auxetic geometries exhibit different crash response behaviors, and (3) optimization via Response Surface Methodology leads to configurations that outperform baseline designs in terms of absorbed energy and deformation control.

2. Methodology

2.1. Filler structure development

In this present study, the filler structure adopts from the double arrowhead auxetic (DAA) model that had highest EA based on research from Chunhao Jiang et al. [29]. The long and short strut angle was constraint to $\theta_1 = 23^\circ$ and $\theta_2 = 68^\circ$, respectively, as presented in Fig. 1. There are two geometric parameters, such as the number of arrowhead arrangement (n) and the thickness of arrowhead struts (t). The numbers of arrowhead arrangement were 2, 3, and 4, that can be seen in Fig. 1

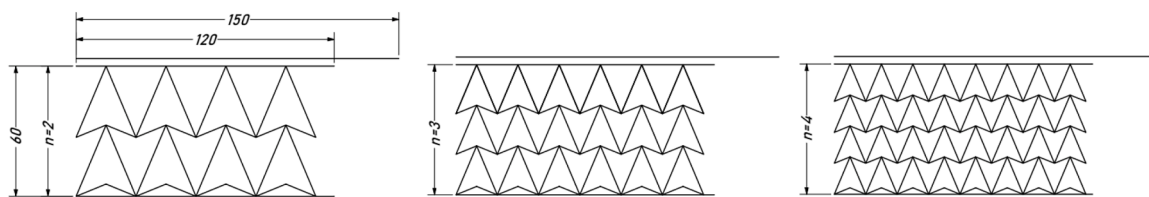


Fig. 1. Geometric configuration of the two-dimensional (2D) Double Arrowhead Auxetic (DAA) structure with different numbers of cell rows: $n=2$, $n=3$, and $n=4$.

and the thicknesses of arrowhead struts were 1, 1.4, and 1.8 mm that can be seen in Fig. 2. The auxetic structure can be constructed in plate lattice and truss lattice. Guo MF et al. modelled the plate-lattice structures with double arrow-head configuration to improve the specific stiffness and energy absorption capability compared to the truss-lattice structure [37]. The result showed that at relatively similar densities, the double arrow-head plate lattice structure produces more stable peak stress and stress-strain curves than the truss-lattice, suggesting potential for better energy absorption applications. So, this study adopts the double arrowhead plate lattice to be applied in ship hull design.

The three-dimensional DAA structure was derived from 2D, it looked like the sandwich panel. G. Sakthi Balan et al., [38] and A. Al Ali et al., [39] used a sandwich configuration because it provides better impact resistance, energy absorption, and structural performance compared to a simple single-layer composite. The arrowhead as the cellular cores was positioned between two rigid plates. Both plates had 60 mm of thickness. The geometric arrangement and simulation setups consisted of a DAA model and an impactor, which was illustrated in Fig. 3. The 3D DAA model and the impactor were built with only $\frac{1}{4}$ geometry. The overall dimensions of the DAA model were 120 mm x 100 mm x 60 mm in length, width, and height, respectively. The impactor had diameter of $D_i = 50$ mm and length of $L_i = 60$ mm, was used to impact the DAA model with an initial velocity of $V = 10$ m/s.

Some previous studies had already used this similar structure with various configurations such as two cell wall angles θ_1 , θ_2 , and length l of the inclined struct between the centerline and the vertex end of arrowhead struct. Matheus et al. [40]. optimized the DAA structure based on meta-modeling. Gao et al. [41]. analyzed the DAA structure with negative Poisson's ratio under axial impact loading. Omid Khaghani et al. [42]. investigated the three different auxetic structures for civil engineering applications, one of which is arrowhead structure, and others were re-entrant and honeycomb structure. These similar studies mentioned that the arrowhead structure could give good mechanical behavior in terms of ultimate force, energy absorption, and specific energy absorption.

2.2. Material properties

For manufacturing this DAA structure, the additive materials such as Polylactic acid (PLA), PLA – Carbon, and AlSi10Mg are considered to be applied. Polylactic acid (PLA) is a material that capable absorb high energy, high crushing force efficiency (CFE) under impact load [43,44]. PLA-Carbon is a composites reinforced with carbon fiber that significantly improve strength-to-weight ratio and stiffness than the pure PLA, as declared by Nabeel Maqsood [45]. PLA – Carbon can enhance the resistance to shear forces, critical in ship hull impacts [46]. The AlSi10Mg is used because this powder feedstock is suitable for lightweight structural and crashworthiness applications [47]. The material of impactor was structural steel and the material of both upper-bottom hull thin walls were set in aluminum alloy. The lightweight aluminum alloy material is recommended to improve the crashworthiness than steel [29]. The mechanical properties of this DAA model and impactor can be seen in Table 1. Investigating various dual-material properties in different type of auxetic structure with NPR had significant effect in compression than in the single material [48].

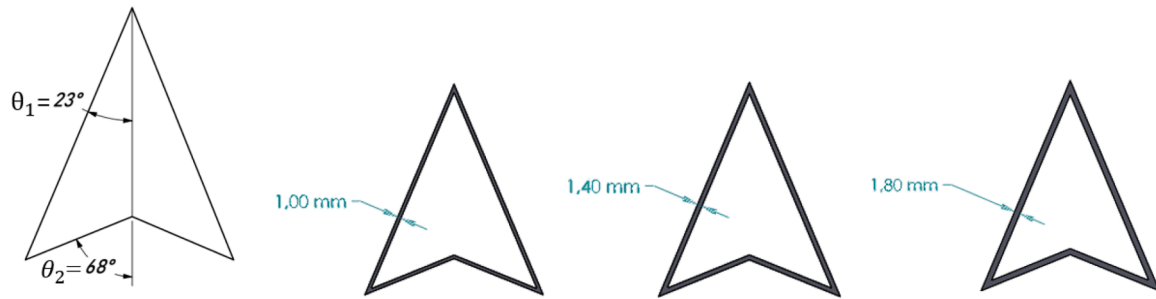


Fig. 2. Geometry of the 2D arrowhead unit cell showing internal angles $\theta_1 = 23^\circ$ and $\theta_2 = 68^\circ$, along with strut thickness $t=1$ mm, $t=1.4$ mm, $t=1.8$ mm.

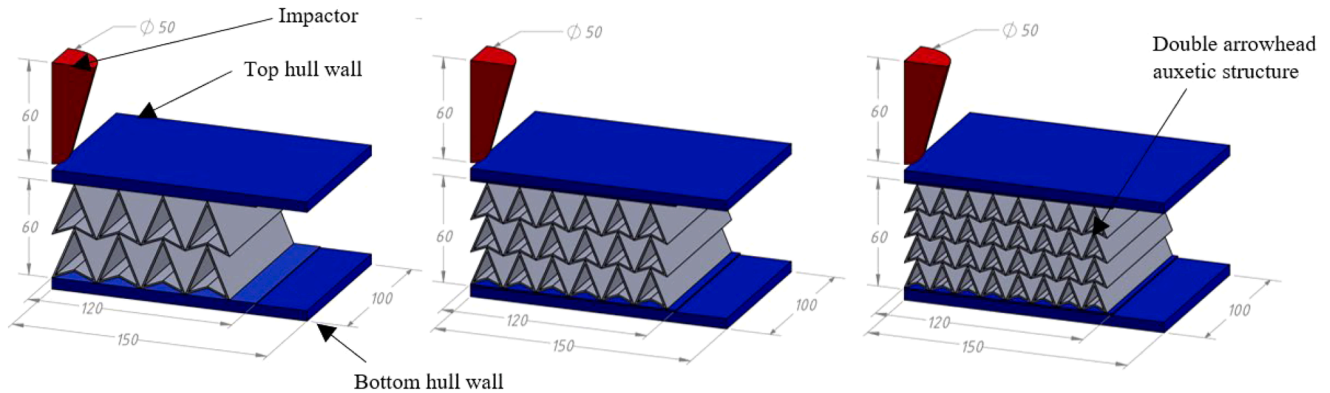


Fig. 3. Three-dimensional (3D) configuration of the drop test setup showing the double arrowhead auxetic (DAA) structure placed between the top and bottom hull walls with Variations in the number of unit cell rows $n = 2$, $n = 3$, and $n = 4$.

Table 1
Mechanical properties of materials used in the simulation model.

Properties	Aluminum Alloy 6063	Structural steel	PLA-Carbon [49]	PLA [50]	AlSi10Mg [51]
Density [kg/m ³]	2270	7850	1190	1250	2640
Modulus elasticity [MPa]	71,000	200,000	29,000	3450	71,000
Poisson's ratio	0.33	0.3	0.38	0.39	0.33
Yield strength [MPa]	280	250	30	54,1	303
Ultimate tensile strength [MPa]	–	460	32.64	–	364
Tangent modulus [MPa]	500	–	380	1E-07	4522
Material applied in	Double hull wall	Impactor	Auxetic	Auxetic	Auxetic

PLA-Carbon was selected for its enhanced energy absorption capabilities, which are crucial for crashworthiness applications. As a filler material in crash boxes, PLA-Carbon significantly improves the energy absorption performance, making it highly effective at mitigating impact forces during collisions [52–54]. PLA also provides a high energy-to-mass ratio, which is valuable for lightweight energy-absorbing applications [55,56]. AlSi10Mg material has ability to form complex optimized structures that can eliminate undesirable failure modes, such as early local collapse, without compromising high energy absorption capability, making it highly suitable for applications

requiring impact resistance, such as crashworthiness systems [57,58].

2.3. Computational simulation

The computational simulation in this study aimed to investigate the drop test performance of the Double Arrowhead Auxetic (DAA) structure using ANSYS explicit dynamic. This module was selected for its ability to accurately simulate high-speed impact events and nonlinear contact interactions, which are critical for evaluating the crashworthiness of auxetic structures. This drop test scenario represented a high-velocity collision, where a rigid impactor collides with the auxetic structure, causing rapid deformation. Due to the high loading rate during impact, the upper cell walls of the auxetic structure interact with adjacent cells, leading to progressive deformation and eventual collapse of the structure's geometry. Initially, the auxetic structure effectively absorbs the kinetic energy due to its lateral deformation capability. However, once the kinetic energy surpasses a certain threshold, the structure reaches its deformation limit, resulting in material failure such as fracturing and collapse. This reduces its ability to absorb and dissipate energy, thereby decreasing its specific energy absorption (SEA) performance [59].

The simulation process was conducted through several stages, as presented in Fig. 4. First, the DAA structure was modeled as a quarter-symmetric 3D surface model. The symmetric assumption helps reduce computational time while maintaining model accuracy [60]. The simulation setup involved a drop-weight rigid impactor and two rigid plates placed at the top and bottom of the structure to ensure consistent impact alignment [61]. Second, material properties were assigned to each component based on experimental and standard data, as detailed in Table 1. The impactor was modeled using structural steel, the top and bottom plates were modeled with aluminum alloy, while the auxetic structure was simulated using three different materials such as PLA-Carbon, PLA, and AlSi10Mg in separate simulation runs to compare their energy absorption capabilities under identical conditions. Third,

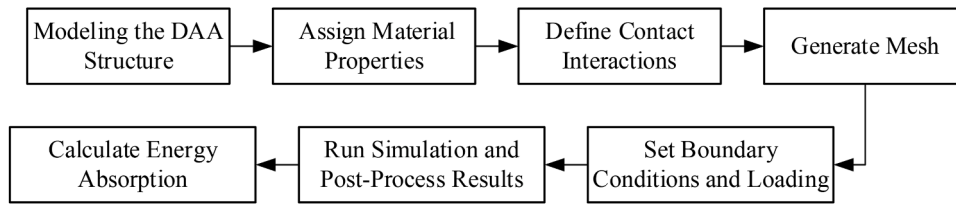


Fig. 4. Flow chart of the numerical simulation process.

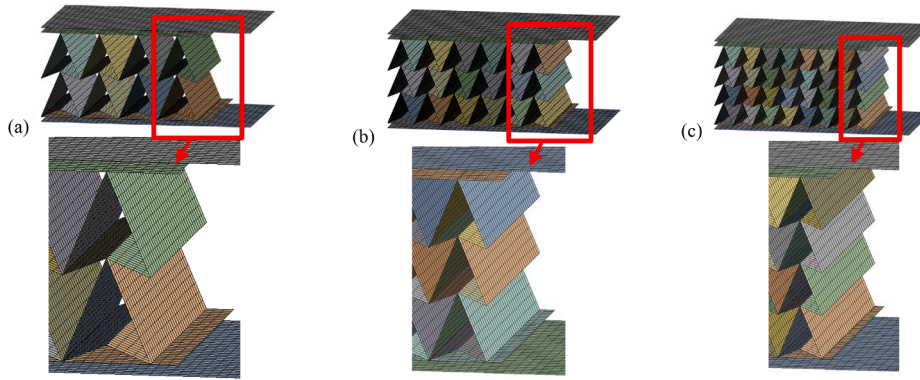


Fig. 5. Finite element mesh generation for auxetic structures with different cell arrangements: (a) two cells ($n = 2$), (b) three cells ($n = 3$), and (c) four cells ($n = 4$). The red box highlights a close-up view of mesh generation.

contact interactions between components were defined as either bonded or frictionless. Bonded contact was used to prevent relative motion between surfaces, while frictionless contact allowed unrestricted sliding between parts. These definitions were essential for accurately simulating the interaction between the auxetic structure and the rigid boundaries [62]. Contact interactions were assigned among components, using bonded contact for areas where separation or sliding was not permitted—such as between the auxetic structure and the rigid plates—while frictionless contact was used for interfaces where unrestricted sliding was allowed, like between the impactor and the top plate. These contact definitions were essential to simulate realistic deformation behavior during the impact event. Fourth, the FE model was evaluated by mesh convergence. The process begins by setting an initial mesh size at 5 mm and refined to 2,5 mm for drop test simulation. Then, the effect on energy absorption was analyzed. The model was converged at a mesh size of 4 mm, as presented in Fig. 5. The mesh quality was evaluated by element quality at Table 2. Fifth, in terms of boundary conditions and contact definitions, the edge of bottom rigid plate was fully fixed in all directions, while the top rigid plate and impactor were allowed to move vertically. For the initial conditions and loading, the impactor was assigned an initial velocity of 10 m/s in the negative y-direction to simulate a free-fall drop impact. The explicit solver was configured with a time step of 5 ms to ensure numerical stability throughout the simulation. Sixth, in order to replicate a realistic impact situation, the simulation was ran until the auxetic structure underwent significant deformation.

The output and post-processing stages involved results such as total deformation, force reaction, and internal energy absorption. Post-processing was performed using contour plots, time-history graphs, and numerical data exported for further analysis. To process the data, simulation outputs such as reaction force and displacement were exported and used to compute energy absorption (EA), which is crucial for assessing the effectiveness of the auxetic structure in impact scenarios. The EA was calculated based on the area under the force-displacement curve obtained from simulations. Three materials such as PLA-Carbon, PLA, and AlSi10Mg, were alternately assigned to the structure to evaluate the effect of material properties on EA performance. It is important to note that this study is limited to numerical

simulations. Experimental validation, including drop test, has been identified as a crucial step and is planned for future work to verify and complement the simulation results presented in this paper.

2.4. Johnson-Cook strength model

The Johnson-Cook (J-C) strength model describes the flow stress (σ) of a material as a function of strain, strain rate, and temperature, and is defined as Eq. (1) [63,64]:

$$\sigma = [A + B(\epsilon)^n] \left[1 - \left(\frac{T - T_r}{T_m - T_r} \right)^m \right] \left[1 + C \ln \left(\frac{\dot{\epsilon}}{\dot{\epsilon}_0} \right) \right] \quad (1)$$

In the equation, ϵ , $\dot{\epsilon}$, and $\dot{\epsilon}_0$ denote the equivalent plastic strain, the plastic strain rate, and the reference strain rate, respectively. T represents the temperature in the deformation zone, T_m is the melting temperature of the material, and T_r refers to room temperature. The constants A , B , n , m , and C correspond to the yield strength, strain hardening coefficient, strain hardening exponent, thermal softening exponent, and strain rate sensitivity coefficient, respectively. In the same case, Rachman Setiawan et al. [65], neglected the temperature effect, the Johnson-Cook constitutive model reduces to the following simplified form:

$$\sigma = [A + B(\epsilon)^n] \left[1 + C \ln \left(\frac{\dot{\epsilon}}{\dot{\epsilon}_0} \right) \right] \quad (2)$$

In this study, the temperature effect in the Johnson-Cook constitutive model is neglected under the assumption of isothermal conditions, as thermal softening was not a dominant factor in the simulated impact scenario. This simplification is currently adopted by Rachman Setiawan

Table 2
Mesh quality of DAA structure with $n = 2$, $n = 3$, $n = 4$.

		$n = 2$	$n = 3$	$n = 4$
Element Quality	Min	0.50851	0.32316	0.50851
	Max	0.99962	0.99987	0.99962
	Average	0.97334	0.97255	0.98202
	Std Dev	6.952e-02	6.7253e-02	5.6252e-02

et al. [65], because the quasi-static test induces deformation that does not generate significant heat, therefore, the thermal effects on the material behavior are considered negligible.

2.5. Energy absorption principle

The simulation in this study was primarily based on energy absorption principle, which focuses on the ability of the structure to absorb and dissipate kinetic energy during an impact event. When the impactor strikes the auxetic structure, its kinetic energy is transferred into the structure and is absorbed through progressive deformation. The energy absorption can be calculated using Eq. (1), where EA is the energy absorption, δ is the maximum displacement, and $F(x)$ is the force function [66,67].

$$EA = \int_0^{\delta} F(x) dx \quad (3)$$

The auxetic design allows for lateral expansion under compression, enabling efficient energy dissipation. Energy absorption (EA) was calculated from the area under the force–displacement curve obtained from the simulation results. This approach provides a quantitative measure of the structure’s crashworthiness, allowing direct comparison of different materials and geometric configurations in terms of their energy absorption capacity. The simulations were performed until significant deformation occurred, capturing the full energy dissipation process. According to Mandar Shinde et al., energy absorption is a crucial parameter for understanding the mechanical characteristics of auxetic structures. In this earlier study, the Ea of six auxetic structures was examined [58]. The greater the energy absorption capacity of a structure, the better it is at absorbing the impact of an impact, which means higher crashworthiness [68].

2.6. Response surface methodology

The Box-Bhenken design would develop the Design of Experiment (DOE) in this study involves three key factors: arrangement number (A), struts thickness (B), and material type (C), each with three levels as presented in Table 3. This three-factor, three-level experimental design allows for the analysis of how these factors interact to optimize energy absorption performance. S.L.C. Ferreira et al., [69] and Ashagrie Mengistu et al., [70] conducted the optimization of analytical methods using Box-Behnken design. By systematically varying these parameters, the study aims to determine the most effective combination for improving crashworthiness in ship structures. The arrangement number represents the number of unit cells in the auxetic structure, with levels set at 2, 3, and 4. A higher arrangement number is expected to influence mechanical behavior by enhancing energy absorption and structural integrity. Struts thickness, measured at 1.0 mm, 1.4 mm, and 1.8 mm, affects the stiffness and strength of the structure, where thicker struts generally provide greater load-bearing capacity and impact resistance. The material factor consists of PLA-Carbon, PLA, and AlSi10Mg, where AlSi10Mg, an aluminum alloy, is anticipated to offer the highest strength and energy absorption, while PLA-Carbon provides a balance between stiffness and flexibility. The response variable in this experiment is the energy absorption (Ea), which is influenced by the interplay between structural configuration, material properties, and geometric thickness. Using statistical techniques such as the Response Surface Method (RSM)

or Factorial Design, the experiment can identify optimal conditions for maximizing energy absorption while maintaining structural efficiency. Matheus et al. applied the Response Surface Methodology (RSM) to achieve optimal design parameters in optimizing double arrowhead unit cells with varying angles and strut lengths [40].

The dataset provided consists of 15 models each defined by variables A, B, and C, with corresponding EA values measured in joules (J), that presented in Table 4.

From Table 4, the combination of $A = 0$, $B = 1$, and $C = 1$ yielded the highest energy absorption at 2049.91 J, while the configuration of $A = -1$, $B = -1$, and $C = 0$ resulted in the lowest at 94.60 J. The effect of strut thickness on energy absorption is corroborated by existing literature. These numerical results have been compared with previous experimental studies, such as those conducted by Saransh Bhardwaj et al. [71], which also examining the mechanical behavior of 3D-printed auxetic structures demonstrated that increasing strut thickness markedly enhances energy absorption capabilities. Specifically, structures with thicker struts exhibited superior performance under compressive loads, attributed to their increased stiffness and strength. As conducted by Anooshe sadat Mirhakimi et al. [72], showed that the unit cell’s topology plays a crucial role in determining energy absorption, especially under high strain rates. In another research by Jianjun Zhang et al. [73], suggested that modifying the geometrical topology can influence the deformation path, enabling the customization of failure modes, stress distribution, load-bearing capacity, and, most importantly, the overall energy absorption of the structure. These findings are consistent with the simulation results, where models with thicker struts achieved the highest energy absorption.

Fig. 6 illustrates the force versus displacement graph of 15 auxetic structure models based on the Design of Experiment (DOE) simulation in Table 4. From the graph, Model 1 and Model 8 exhibit the highest peak forces, indicating that these configurations have the strongest resistance to deformation. These models likely feature optimal parameters, such as thicker struts ($B = 1$), stronger material ($C = 1$), or a well-balanced arrangement number ($A = 0$ or $A = 1$). Their curves show a steady increase in force with displacement, suggesting a high energy absorption capacity. On the other hand, Models 3, 7, and 12 demonstrate moderate force responses, indicating a balanced combination of material properties and structural configuration. These models likely incorporate mid-range strut thickness or different material compositions, such as PLA-based carbon or PLA. Conversely, Models 10, 11, and 14 exhibit the lowest force values, remaining below 20,000 N throughout the deformation process. These models likely have thinner struts ($B = -1$) and weaker materials ($C = 0$ or $C = -1$), leading to reduced impact resistance and lower energy absorption. The curves for these models show early force saturation or irregular behavior, indicating premature failure or lower structural stiffness.

The simulation results presented in Fig. 6 align well with the experimental and simulation outcomes by Rachman Setiawan et al. [65], particularly for the double arrowhead auxetic model. In Fig. 6, several models demonstrate a progressive increase in force with displacement, followed by fluctuations that indicate structural deformation and energy absorption during impact loading in auxetic structures. Fig. 6 also compared to research from Qiang Gao et al. [74], that the 3D Double Arrowhead Auxetic (DAA) structure under quasi-static compression loading exhibits superior and validated mechanical behavior, with higher load capacity, stable deformation, and excellent agreement between simulation and experiment, thereby demonstrating the same relationship observed in the simulation results of Fig. 6. This coherence confirms that the simulation methodology applied in Fig. 6 is reliable, as it produces results that are experimentally validated in previous research.

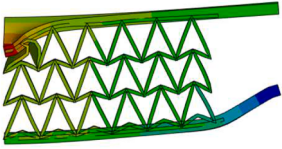
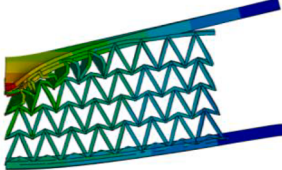
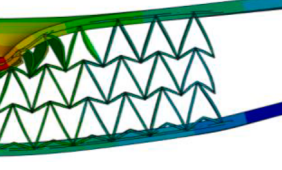
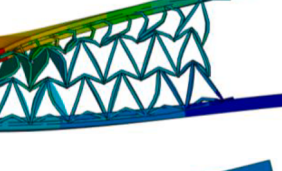
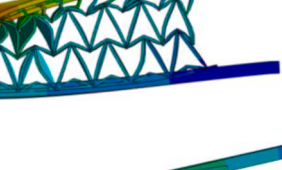
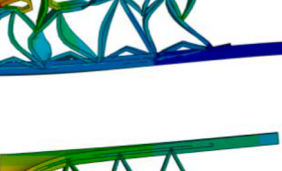
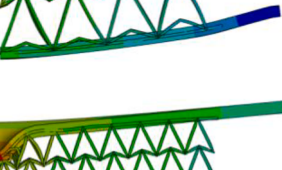
Material selection also plays a crucial role in determining energy absorption efficiency. Stefan Bronder et al. [75], investigated the hybrid auxetic structures has revealed that materials with higher strength and stiffness, such as AlSi10Mg, significantly improve energy absorption

Table 3

The parameters used in Design of experiment (DOE).

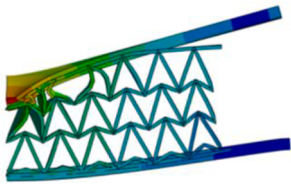
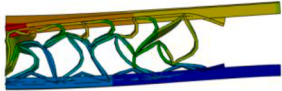
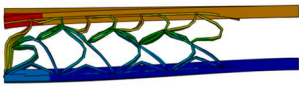
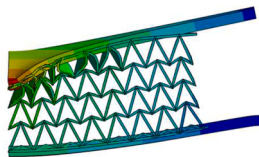
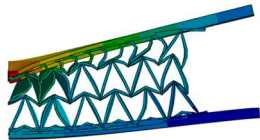
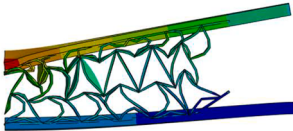
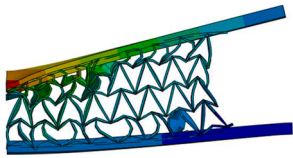
Factor	Symbol	Level –1	Level 0	Level 1
arrangement number	A	2	3	4
struts thickness	B	1	1,4	1,8
material	C	PLA-Carbon	PLA	AlSi10Mg

Table 4
The DOE of Box-Bhenken design.

Model	Variables			Response	
	A	B	C	EA (J)	Deformation illustration
1	0	1	1	2049.91	Total Deformation Type: Total Deformation Unit: mm Time: 5.e-003 s Cycle Number: 26330 02/09/2024 10:06 50 Max 44.444 38.889 33.333 27.778 22.222 16.667 11.111 5.5556 0 Min 
2	1	1	0	584.31	Total Deformation Type: Total Deformation Unit: mm Time: 5.0001e-003 s Cycle Number: 20677 07/09/2024 08:53 50.001 Max 44.445 38.889 33.334 27.778 22.222 16.667 11.111 5.5556 0 Min 
3	0	-1	1	1023.99	Total Deformation Type: Total Deformation Unit: mm Time: 5.e-003 s Cycle Number: 26330 02/09/2024 10:25 50 Max 44.444 38.889 33.333 27.778 22.222 16.667 11.111 5.5556 0 Min 
4	0	0	0	371.80	Total Deformation Type: Total Deformation Unit: mm Time: 5.e-003 s Cycle Number: 26330 02/09/2024 11:13 50 Max 44.444 38.889 33.333 27.778 22.222 16.667 11.111 5.5556 0 Min 
5	0	0	0	371.80	Total Deformation Type: Total Deformation Unit: mm Time: 5.e-003 s Cycle Number: 26330 02/09/2024 12:31 50 Max 44.444 38.889 33.333 27.778 22.222 16.667 11.111 5.5556 0 Min 
6	-1	1	0	301.38	Total Deformation Type: Total Deformation Unit: mm Time: 5.0002e-003 s Cycle Number: 7291 07/09/2024 09:44 50.002 Max 44.446 38.891 33.335 27.779 22.223 16.667 11.112 5.5558 0 Min 
7	-1	0	1	1057.33	Total Deformation Type: Total Deformation Unit: mm Time: 5.0001e-003 s Cycle Number: 20677 02/09/2024 17:34 50.001 Max 44.445 38.889 33.334 27.778 22.222 16.667 11.111 5.5556 0 Min 
8	1	0	1	1875.16	Total Deformation Type: Total Deformation Unit: mm Time: 5.0001e-003 s Cycle Number: 20677 02/09/2024 19:44 50.001 Max 44.445 38.889 33.334 27.778 22.222 16.667 11.111 5.5556 0 Min 

(continued on next page)

Table 4 (continued)

Model	Variables			Response	
	A	B	C	EA (J)	Deformation illustration
9	0	1	−1	289.89	Total Deformation Type: Total Deformation Unit: mm Time: 5.e+003 s Cycle Number: 26330 01/06/2024 13.22 50 Max 44.444 33.333 27.778 22.222 16.667 11.111 5.5556 0 Min 
10	−1	0	−1	171.56	Total Deformation Type: Total Deformation Unit: mm Time: 5.0001e+003 s Cycle Number: 19677 14/04/2025 13.39 50.001 Max 44.443 33.339 27.776 22.223 16.667 11.111 5.5556 0 Min 
11	−1	−1	0	94.60	Total Deformation Type: Total Deformation Unit: mm Time: 5.0001e+003 s Cycle Number: 19677 14/04/2025 14.57 50.001 Max 44.443 33.334 27.776 22.223 16.667 11.111 5.5556 0 Min 
12	1	0	−1	573.10	Total Deformation Type: Total Deformation Unit: mm Time: 5.0001e+003 s Cycle Number: 19677 14/04/2025 15.36 50.001 Max 44.443 33.334 27.776 22.223 16.667 11.111 5.5556 0 Min 
13	0	0	0	371.80	Total Deformation Type: Total Deformation Unit: mm Time: 5.e+003 s Cycle Number: 26330 14/04/2025 14.38 50 Max 44.444 33.333 27.778 22.222 16.667 11.111 5.5556 0 Min 
14	0	−1	−1	201.97	Total Deformation Type: Total Deformation Unit: mm Time: 5.e+003 s Cycle Number: 26330 14/04/2025 15.23 50 Max 44.444 33.333 27.778 22.222 16.667 11.111 5.5556 0 Min 
15	1	−1	0	362.17	Total Deformation Type: Total Deformation Unit: mm Time: 5.0001e+003 s Cycle Number: 19677 14/04/2025 14.23 50.001 Max 44.443 33.334 27.776 22.223 16.667 11.111 5.5556 0 Min 

compared to their polymer-based counterparts. Stefan Bronder et al. [75]. found that incorporating materials like AlSi10Mg into auxetic designs enhances their mechanical properties, leading to better performance under impact conditions. This is consistent with the simulation results, where the use of AlSi10Mg material corresponded with superior energy absorption outcomes. It indicates that the applying the better mechanical properties to DAA model, the Ea value increases [76].

Furthermore, the arrangement number, indicative of the geometric configuration or the number of unit cells in the structure, has a substantial impact on energy absorption. An increased arrangement number contributes to a more complex structural configuration, which effectively distributes impact forces and enhances energy absorption. This observation is supported by A. C. Kaya et al. that emphasizing the importance of structural design in optimizing the mechanical performance of auxetic materials [77].

The Fig. 7 presents 3D response surface plots illustrating the

relationship between energy absorption (EA) and different design parameters: arrangement, thickness, and material. Those figure shows the surface gradient increasing by the increase of parameters. The thickness increases, EA also increases, while arrangement has a varying effect. The thicker structures indicate the more energy absorbed. The material with high mechanical properties plays crucial role in responding the deformation behavior. The results indicate that the choice of parameters plays a crucial role in determining the energy absorption capacity of the Double Arrowhead Auxetic (DAA) structure. The highest energy absorption is achieved when using an optimal combination of factors, while suboptimal parameter settings result in significantly lower energy absorption. These findings provide valuable insights for optimizing the DAA structure to enhance its crashworthiness and impact resistance.

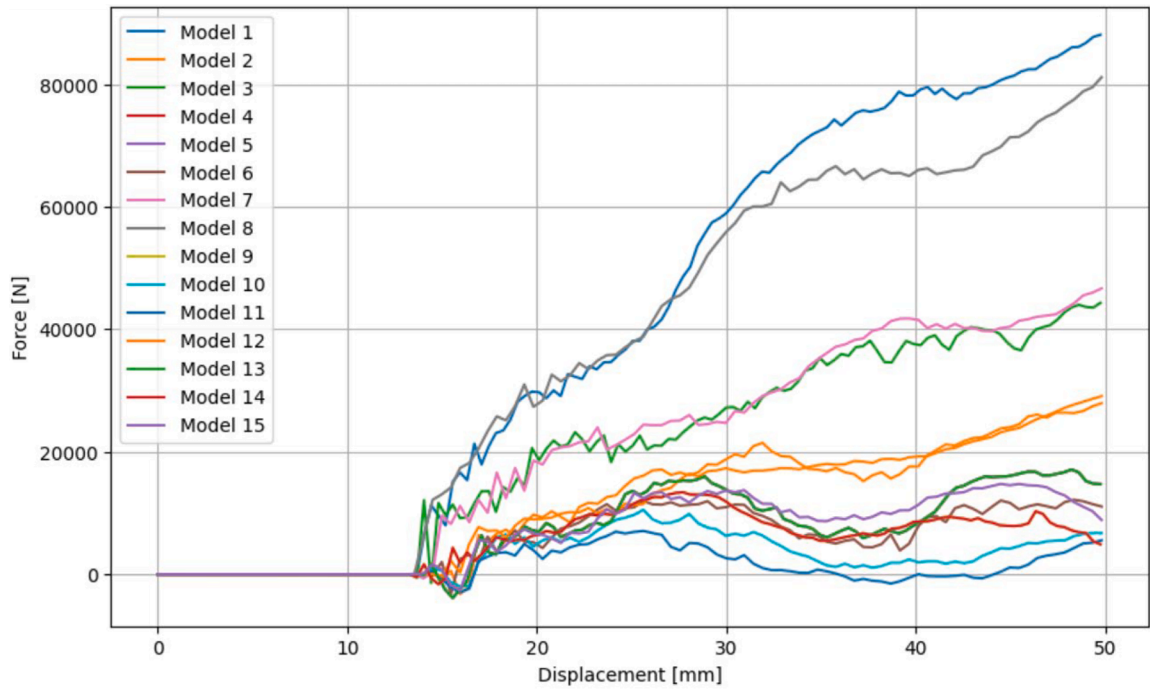


Fig. 6. Comparison of force versus displacement for 15 different structural models under drop test simulation.

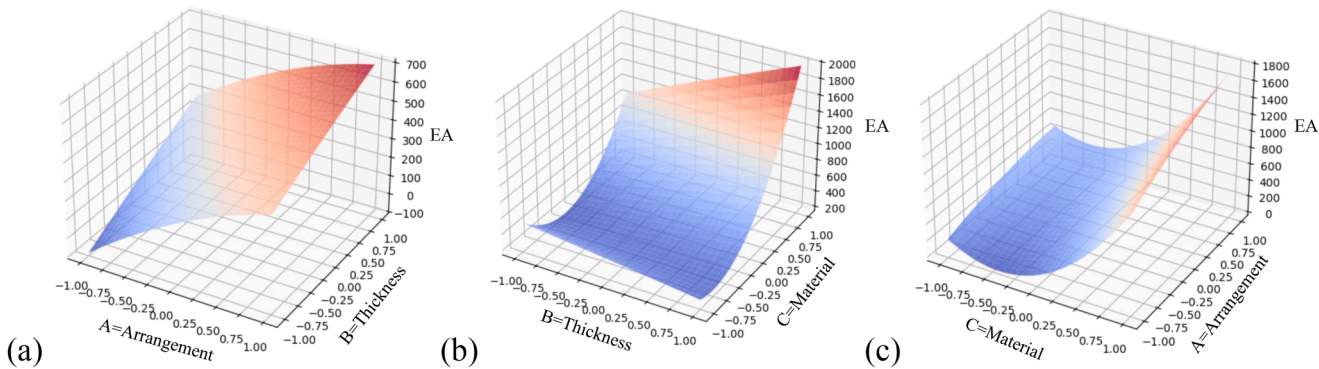


Fig. 7. 3D plot surface of (a) Arrangement and thickness, (b) thickness and material, (c) material and arrangement.

2.7. Regression analysis

In this study, regression analysis was employed to verify the relationship between the energy absorption (EA) as the response variable and several key independent variables, including wall thickness, material type, and number of auxetic cells. The regression equation is typically expressed as:

$$EA = \beta_0 + \sum \beta_i X_i + \sum \beta_{ii} X_i^2 + \sum \beta_{ij} X_i X_j + \epsilon$$
 (4)

Where β_0 is the intercept, β_i , β_{ii} , β_{ij} are the regression coefficients, and ϵ is the error term. The quality and validity of the regression model were evaluated using p-value that helps us determine whether an independent variable (or predictor) has a statistically significant effect on the dependent variable (in this case, Energy Absorption, or EA).

The regression model for response EA was developed using polynomial regression analysis to estimate the effects of factors A, B, and C, along with their quadratic and interaction terms. As demonstrated by Parsa Behinfar et al., who optimize flexural stiffness, axial and radial elasticity, and Poisson's ratio by establishing a polynomial relationship between stent shape and elasticity parameters, the findings are

confirmed by analytical modeling [68]. Jiyun Kim et al. used this method for effectively capturing the relationship between variables and responses with limited data for analysis and optimization [78]. Table 5 presents the estimated regression coefficients for response variable EA. The constant term has a coefficient of 371.800, representing the baseline

Table 5
Estimated Regression Coefficients for E.

Response	Coefficient	Standard Error of Coefficient	T-value	P-value
Constant	371.800	94.51	3.934	0.011
A	221.234	57.88	3.823	0.012
B	192.845	57.88	3.332	0.021
C	596.234	57.88	10.302	0.000
A ²	-4.169	85.19	-0.049	0.963
B ²	-32.016	85.19	-0.376	0.722
C ²	551.656	85.19	6.475	0.001
AB	3.840	81.85	0.047	0.964
AC	104.073	81.85	1.272	0.259
BC	234.500	81.85	2.865	0.035

S (Standard Error) = 163.700; PRESS = 2143,808.
R-Squared (R²) = 97.36 %; R-Sq(Predicted) = 57.81 %; R-Sq(Adjusted) = 92.62 %.

response when all factors are at their reference levels. The linear terms for A , B , and C have positive coefficients, indicating that increasing these factors generally increases EA , with factor C having the most significant impact (596.234). The quadratic terms exhibit varying effects; while A^2 and B^2 show weak influence with high p -values, C^2 has a strong and significant effect ($p = 0.001$), highlighting a nonlinear relationship. Interaction effects vary, with BC showing a significant interaction ($p = 0.035$), whereas AB and AC do not contribute significantly to the model. The R-squared value ($R-Sq = 97.36\%$) indicates that 97.36 % of the variation in the response variable can be explained by the regression model. This means that the model fits the training data very well, capturing almost all of the variability in the observed values. The large gap between Adjusted R^2 and Predicted R^2 suggests that the model fits the training data very well but lacks generalizability or over fitting that caused by retaining too many non-significant variables (such as A^2 , B^2 , AB , and AC), which contribute little to the predictive power but increase model complexity. In order to decrease overfitting and increase predicted accuracy, only the statistically relevant variables (A , B , C , C^2 , and BC) should be included. This would simplify the model and potentially lead to a better balance between goodness-of-fit and predictive performance.

From tabel 5, the regression model can be built as the foundation for response surface optimization, where the function to be optimized is expressed in Eq. (5). This equation represents the response function that will be used to determine optimal conditions in the problem under study. Majid Lotfi et al. declared that the optimization using the response surface method (RSM) confirmed the proposed structure's superior energy absorption compared to conventional designs [79].

$$R = 371.800 + 221.234A + 192.845B + 596.234C - 4.169A^2 - 32.016B^2 + 551.656C^2 - 3.840AB + 104.073AC + 234.500BC \quad (5)$$

The sensitivity analysis conducted through the regression model in this study reveals that material type (factor C) has the most significant influence on EA , both in terms of its linear and quadratic effects. This indicates that changes in the material used can lead to substantial variations in the energy absorption performance of the auxetic structure. In comparison, arrangement (A) and thickness (B) also contribute to EA , but their impact is relatively less pronounced than that of the material. Moreover, the interaction between factor B and factor C was found to be statistically significant, suggesting that the effectiveness of a particular material may depend on the cells arrangement used. This interaction highlights the importance of considering combined effects, as material performance may vary under different structural configurations.

The normal probability plot shown in Fig. 8 presents the normal probability plot of the residuals for the energy absorption (EA) response that shows the residuals closely follow a straight diagonal line, suggesting that the residuals are approximately normally distributed. This indicates that the assumption of normality is satisfied, which is crucial for ensuring the validity of statistical inferences such as p -values and confidence intervals in the regression model.

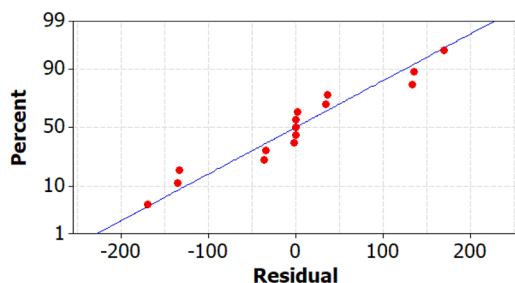


Fig. 8. Normal probability plot of residuals used to assess the normality assumption in the regression analysis.

The regression model's projected response values for several new design points are shown in Table 6. For instance, Model 1 has the highest predicted response, suggesting that this design configuration provides the most effective energy absorption. In contrast, Model 11 has the lowest predicted response, indicating that this configuration is not suitable for improving energy absorption performance. The optimal design is predicted for maximizing energy absorption can be seen in Table 7. Those mean that the design constructed in 4 strut arrangements, 1,8 mm of thickness, and AlSi10Mg of material selection.

3. Results

3.1. Response surface methodology analysis

The optimization design is analyzed numerically in the drop test simulation. The deformation result shows in each 10 mm interval of impactor movement and compares to the non-optimized model, as presented in Fig. 9.

As shown in Fig. 9, the failure mechanisms of the optimized and non-optimized lattice structures under drop test simulation demonstrate clear differences in structural performance. In the optimized structure (1–5), the response begins with an undeformed, uniform triangular lattice designed to distribute stress progressively. As loading initiates (point 1–2), the structure undergoes elastic deformation with minimal stress due to its compliant design. At point 2–3, localized stress concentrations emerge, particularly near the top corners, marking the onset of local buckling. Despite this, the structure continues to transfer load efficiently, as indicated by a sharp rise in stress. Progressing to point 3–4, more members begin to buckle, yet the structure maintains its integrity. The deformation becomes more visible, especially around high-stress zones, but failure remains localized and controlled. Finally, at point 4–5, significant structural failure occurs in the top cells, with concentrated buckling and possible ruptures. Nonetheless, the structure continues to bear substantial load, showcasing a resilient and gradual failure mechanism due to its optimized geometry, which enables better energy absorption and delayed collapse. One of the most notable features observed during the failure process is the shrinkage behavior of the optimized DAA model under compressive loading. As the structure is compressed by the impactor, the unit cells begin to collapse inward toward the center of the model. This behavior is characteristic of a negative Poisson's ratio (NPR), where lateral dimensions contract instead of expanding when compressed [80]. The NPR effect is clearly evident in the deformation stages (3–5), where the collapsing mechanism results in a more compact structure, enhancing the overall stiffness and resistance to further deformation. This internal compaction not only improves structural stability but also contributes significantly to energy absorption, as more load paths are activated during the densification phase. The auxetic behavior allows the structure to maintain

Table 6
Predicted Response for New Design Points Using Model for E_a .

Model	Fit	SE Fit	95 % CI	95 % PI
1	1915.02	141.768	(1550.59; 2279.45)	(1358.35; 2471.69)
2	753.53	141.768	(389.11; 1117.96)	(196.86; 1310.20)
3	1060.33	141.768	(695.90; 1424.76)	(503.66; 1617.00)
4	371.80	94.512	(128.85; 614.75)	(-114.10; 857.70)
5	371.80	94.512	(128.85; 614.75)	(-114.10; 857.70)
6	303.39	141.768	(-61.04; 667.81)	(-253.28; 860.06)
7	1190.21	141.768	(825.79; 1554.64)	(633.54; 1746.89)
8	1840.83	141.768	(1476.40; 2205.25)	(1284.16; 2397.50)
9	253.55	141.768	(-110.88; 617.98)	(-303.12; 810.22)
10	205.89	141.768	(-158.53; 570.32)	(-350.78; 762.56)
11	-74.62	141.768	(-439.05; 289.80)	(-631.29; 482.05)
12	440.22	141.768	(75.79; 804.64)	(-116.46; 996.89)
13	371.80	94.512	(128.85; 614.75)	(-114.10; 857.70)
14	336.86	141.768	(-27.57; 701.29)	(-219.81; 893.53)
15	360.16	141.768	(-4.26; 724.59)	(-196.51; 916.83)

Table 7
The optimum configuration.

Configuration	Level	Value
Arrangement (A)	1	4
Thickness (B)	1	1.8
Material (C)	1	AlSi10Mg
Energy Absorption (EA)		

load-bearing capacity even at higher strains, preventing premature failure.

In contrast, the non-optimized structure (i-v) displays a much less efficient failure response. Initially, it begins undeformed (i) but lacks the optimized geometric design needed for gradual load transfer. At an early stage (ii), localized buckling starts prematurely with minimal stress development. As deformation progresses (iii), the buckling quickly spreads across the structure. By point iv, severe bending and deformation of structural elements occur, yet the maximum stress reached is still significantly lower than that of the optimized model. Finally, at point v, the structure undergoes catastrophic failure. The non-optimized design, lacking geometrical refinement and the NPR mechanism, fails to distribute stress effectively, resulting in early, unstable buckling and poor energy dissipation.

Overall, the optimized model clearly demonstrates superior mechanical behavior through staged failure and internal densification. Its ability to exhibit a negative Poisson's ratio under impact loading leads to delayed collapse, better energy absorption, and improved structural resilience compared to the non-optimized structure. Todd Shepherd et al. had proved by experiment test and simulation analysis that auxetic lattices demonstrated superior energy dispersion and were more resistant to bottoming out, due to their characteristic lateral expansion during compression [81]. Compared to traditional metal auxetic structures, the hybrid double arrowhead auxetic (DAA) design absorbed 56.4 % more energy and exhibited a lower peak crushing force [82]. Local impact tests were done by Weiwei Jiang et al. to study how auxetic structures behave under sudden loads. The results showed that structures with higher relative density experienced less deformation and damage, making them more impact-resistant. The study also found that different impulse levels significantly affected the structures' responses,

indicating that their performance can be customized for specific impact conditions [83]. The Double Arrowhead Auxetic (DAA) model was also compared with non-auxetic structures, specifically the honeycomb model reported by Jin Zhou et al. [14]. It achieved up to 232 % higher initial compressive strength and 125 % greater specific energy absorption (SEA). Omid Khaghani et al. [42]. reported that the DAA model absorbed 176.8 % more energy than the honeycomb structure. Similarly, Mohsen Safikhani Nasim et al. [84]. conducted a comprehensive comparison under quasi-static compression, showing that the total energy absorbed by the re-entrant auxetic structure was 57.75 % higher, with a 26.19 % increase in SEA, compared to the non-auxetic honeycomb.

4. Discussion

This study investigates the enhancement of crashworthiness in double-walled ship hulls through the integration of double arrowhead auxetic (DAA) structures as a filler structure in void space. The findings indicated that DAA structures significantly improve energy absorption capabilities, thereby augmenting the hull's ability to withstand impact forces. The optimization process using Response Surface Methodology (RSM) identified the most effective combination of design parameters, with AlSi10Mg material, a strut thickness of 1.8 mm, and a four-arrowhead arrangement yielding the highest energy absorption. This outcome is consistent with previous research, such as the work by Q. Gao et al. which demonstrated that auxetic structures with double arrowhead truss lattice geometries exhibit enhanced impact resistance and energy dissipation [41]. Similarly, Jianhua Donget al. reported that specific auxetic configurations could effectively dissipate energy under dynamic loading conditions [85]. The study further explored the impact of geometric parameters, specifically cell thickness and the number of arrowhead arrangements, on energy distribution. Findings indicate that increasing the strut thickness and the number of arrowheads enhances the structure's ability to distribute and absorb energy. This observation aligns with the research conducted by Wang et al. who found that augmenting the scale factor of auxetic structures leads to improvements in their elastic modulus and impact performance [37]. Additionally, Yazıcı, Mustafa Enes et al. (2024) demonstrated that modifications in geometric configurations of auxetic materials could result in superior

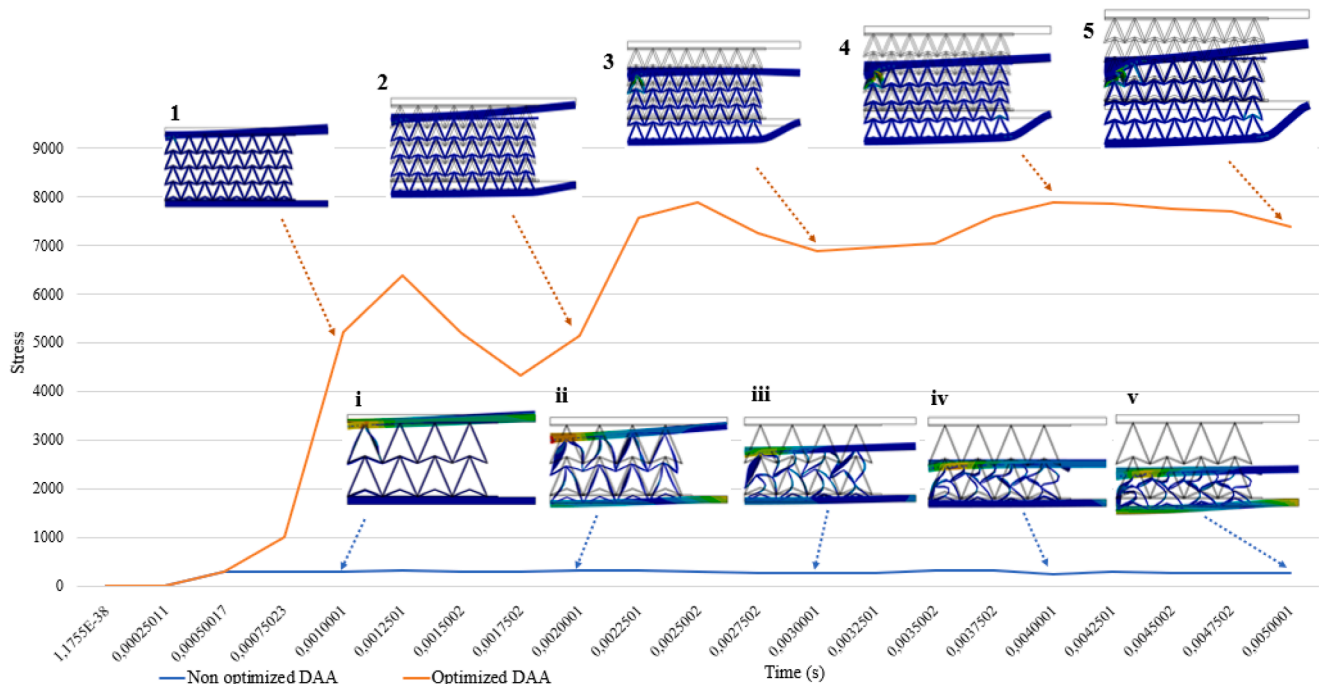


Fig. 9. The failure mechanisms comparison of optimized (1–5) dan non-optimized (i-v) model.

mechanical properties, including increased energy absorption [86]. However, it is important to note that excessive thickness may lead to a reduction in flexibility and an increase in overall weight, which could be detrimental in certain applications.

From a mechanical perspective, the negative Poisson's ratio property of the DAA structure played a crucial role in improving impact resistance. The NPR effect allowed the structure to undergo controlled deformation under impact loads, reducing stress concentrations and preventing catastrophic failure. This behavior has been extensively studied by Jianjun Zhang et al. who demonstrated that auxetic materials exhibit superior energy absorption characteristics compared to conventional materials with positive Poisson's ratios [73]. The findings of this study further support the hypothesis that auxetic structures can be effectively integrated into ship hull designs to enhance their crashworthiness.

Material selection emerged as a critical factor influencing energy absorption performance. The study identified AlSi10Mg as the material yielding the highest energy absorption. This finding is corroborated by Hiromi Yasuda et al. (2021), who observed that AlSi10Mg-based origami cellular structures exhibit superior strength and ductility under impact loading [86]. In contrast, the use of PLA-based materials, particularly PLA-Carbon, demonstrated moderate performance in energy dissipation. This observation is in line with the studies by Davide Mocerino et al. [87], which highlighted that while PLA-based composites offer certain advantages, their energy absorption capacity is generally lower compared to metal-based auxetic structures.

The response surface analysis conducted in this study validated the interactions between geometric and material parameters, confirming their collective influence on energy absorption performance. This analytical approach is supported by the methodologies employed by Matheus Brendon Francisco et al. who successfully utilized statistical models to optimize auxetic structures for improved mechanical performance [40]. This research highlighted the importance of considering both material properties and geometric configurations in the design of auxetic structures to achieve desired energy absorption outcomes.

Overall, the results confirm that the integration of optimized DAA structures into double-walled ship hulls significantly enhances their impact resistance and energy absorption capacity. The study bridges the gap in maritime safety research by providing a systematic approach for designing auxetic fillers that improve structural integrity during high-speed marine collisions. Future work should focus on experimental validation through physical drop tests and large-scale impact simulations to further refine the applicability of these optimized structures in real-world maritime environments. Additionally, further research into hybrid auxetic materials could provide insights into the development of lightweight yet high-strength solutions for crashworthy ship hull designs.

5. Conclusions

This study demonstrates the effectiveness of Double Arrowhead Auxetic (DAA) structures in enhancing the crashworthiness of double-walled ship hulls. By employing Response Surface Methodology (RSM), the optimal configuration for maximizing energy absorption was identified, with the best performance observed in AlSi10Mg material, a strut thickness of 1.8 mm, and a four-arrowhead arrangement. Specifically, the DAA structure showed superior energy absorption capability compared to conventional non-auxetic designs such as honeycomb geometry. This optimal DAA configuration demonstrated the most efficient deformation mechanism under impact. This means the structure can absorb collision energy more effectively and distribute it more uniformly, helping to prevent severe structural damage to the ship's hull. This structure is recommended to be placed in the void space between the two layers of a double-walled hull. This region is highly vulnerable to external impacts, such as during ship-to-ship collisions. Future work should focus on experimental validation and real-scale applications to

further refine the structural performance under dynamic loading conditions.

CRedit authorship contribution statement

Moch. Agus Choiron: Writing – review & editing, Writing – original draft, Supervision, Methodology, Conceptualization. **Yudy Surya Irawan:** Writing – review & editing, Writing – original draft, Supervision, Methodology, Conceptualization. **Rosadila Febritasari:** Writing – review & editing, Writing – original draft, Methodology. **Johan Wayan Dika:** Writing – review & editing, Writing – original draft, Methodology.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests:

Moch. Agus Choiron reports financial support was provided by Directorate of Research and Community Service (DRPM) Brawijaya University. Moch. Agus Choiron reports a relationship with Brawijaya University that includes: employment. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Raw data of DAA performance, method, software, hardware (Original data) (Mendeley Data).

References

- [1] Q. Liu, Materials with negative Poisson's ratios and potential applications to aerospace and defence, *Lit. Rev.* (2006) 1–31 [Online]. Available, <http://oai.dtic.mil/oai?verb=getRecord&metadataPrefix=html&identifier=ADA460791>.
- [2] Y. Zhou, Y. chen Xie, T. Pan, W. Zhu, H. Zhang, G. yan Huang, Flexible materials and structures for mitigating combined blast and fragment loadings—A review, *Int. J. Impact Eng.* 181 (2023) 104759, <https://doi.org/10.1016/j.ijimpeng.2023.104759> no. August.
- [3] W. Jiang, et al., Manufacturing, characteristics and applications of auxetic foams: a state-of-the-art review, *Compos. B Eng.* 235 (2022) 109733, <https://doi.org/10.1016/j.compositesb.2022.109733> no. September 2021.
- [4] I.I. Argatov, R. Guinovart-Díaz, F.J. Sabina, On local indentation and impact compliance of isotropic auxetic materials from the continuum mechanics viewpoint, *Int. J. Eng. Sci.* 54 (2012) 42–57, <https://doi.org/10.1016/j.jengsci.2012.01.010>.
- [5] N. Novak, M. Krstulović-Opara, Z. Ren, M. Vesnjak, Compression and shear behaviour of graded chiral auxetic structures, *Mech. Mater.* 148 (2020), <https://doi.org/10.1016/j.mechmat.2020.103524> no. March.
- [6] J. Blachut, Buckling behaviour of auxetic domes under external pressure, *Thin-Walled Struct.* 182 (2023) 110262, <https://doi.org/10.1016/j.tws.2022.110262> no. September 2022.
- [7] B. Necemer, Z. Božić, S. Glodež, Fatigue resistance of the auxetic honeycombs, *Procedia Struct. Integr.* 46 (2021) (2023) 68–73, <https://doi.org/10.1016/j.prostr.2023.06.012>.
- [8] E. Etemadi, M. Gholikord, M. Zeeshan, H. Hu, Improved mechanical characteristics of new auxetic structures based on stretch-dominated-mechanism deformation under compressive and tensile loadings, *Thin-Walled Struct.* 184 (2023) 110491, <https://doi.org/10.1016/j.tws.2022.110491> no. December 2022.
- [9] M. Najafi, H. Ahmadi, G. Liaghat, Experimental investigation on energy absorption of auxetic structures, *Mater., Today Proc.* 34 (2019) 350–355, <https://doi.org/10.1016/j.matpr.2020.06.075>.
- [10] T. Gärtner, S.J. van den Boom, J. Weerheijm, L.J. Sluys, Geometric effects on impact mitigation in architected auxetic metamaterials, *Mech., Mater.* 191 (2024) 104952, <https://doi.org/10.1016/j.mechmat.2024.104952> no. February.

- [11] F. Scarpa, L.G. Ciffo, J.R. Yates, Dynamic properties of high structural integrity auxetic open cell foam, *Smart Mater., Struct.* 13 (49) (2003), <https://doi.org/10.1088/0964-1726/13/1/006>.
- [12] X. Lan, L. Meng, J. Zhao, Z. Wang, Mechanical properties and damage characterizations of 3D double-arrowhead auxetic structure with high-relative-density realized via selective laser melting, *Eur. J. Mech. A/Solids* 90 (2021) 104386, <https://doi.org/10.1016/j.euromechsol.2021.104386> no. July.
- [13] B. Su, et al., Multiaxial yield behavior of 2D re-entrant auxetic cellular materials, *Eng., Struct.* 311 (2024) 118216, <https://doi.org/10.1016/j.engstruct.2024.118216> no. January.
- [14] J. Zhou, H. Liu, J.P. Dear, B.G. Falzon, Z. Kazanci, Comparison of different quasi-static loading conditions of additively manufactured composite hexagonal and auxetic cellular structures, *Int. J. Mech. Sci.* 244 (2023) 108054, <https://doi.org/10.1016/j.ijmecsci.2022.108054> no. November 2022.
- [15] J.X.C.Q.C. Qiao, Impact resistance of uniform and functionally graded auxetic double arrowhead honeycombs, *Int. J. Impact Eng.* 83 (2015), <https://doi.org/10.1016/j.ijimpeng.2015.04.005>.
- [16] Q. Gao, W.H. Liao, L. Wang, On the low-velocity impact responses of auxetic double arrowed honeycomb, *Aerosp., Sci., Technol.* 98 (2020) 105698, <https://doi.org/10.1016/j.ast.2020.105698>.
- [17] L. Gu, Q. Xu, D. Zheng, H. Zou, Z. Liu, Z. Du, Analysis of the mechanical properties of double arrowhead auxetic metamaterials under tension, *Text., Res., J.* 90 (21–22) (2020) 2411–2427, <https://doi.org/10.1177/0040517520924850>.
- [18] A. Mohan, P. Mondal, J. Rengaswamy, Impact behavior of auxetic structures: experimental and numerical analysis, *Mater. Today Proc* 87 (2023) 292–298, <https://doi.org/10.1016/j.matpr.2023.05.631>.
- [19] S. Shukla, B.K. Behera, Auxetic fibrous structures and their composites: a review, *Compos., Struct.* 290 (2022) 115530, <https://doi.org/10.1016/j.compstruct.2022.115530> no. March.
- [20] B. Lin, X. Dong, Ship hull inspection: a survey, *Ocean Eng* 289 (2023), <https://doi.org/10.1016/j.oceaneng.2023.116281> no. November.
- [21] S. P. and G.K. Aggelos N. Pilatis, Dimitrios-Nikolaos Pagonis, Michael Serris, “A statistical analysis of ship accidents (1990–2020) focusing on collision, grounding, hull failure, and resulting hull damage,” no. January 2024, [doi:10.3390/jmse12010122](https://doi.org/10.3390/jmse12010122).
- [22] J. Magliaro, W. Altenhof, Mechanical performance and crashworthiness of plates and extrusions subjected to cutting: an overview, *Thin-Walled Struct* 148 (2020) 106612, <https://doi.org/10.1016/j.tws.2020.106612> no. December 2019.
- [23] Z. Xiong, M. Ma, Y. Guo, R. Li, Study of energy absorption characteristics of coastal ship side structures based on collision accidents, *Ocean Eng* 307 (2024), <https://doi.org/10.1016/j.oceaneng.2024.118079> no. January.
- [24] B. Liu, C.Guedes Soares, Assessment of the strength of double-hull tanker side structures in minor ship collisions, *Eng., Struct.* 120 (2016) 1–12, <https://doi.org/10.1016/j.engstruct.2016.04.011>.
- [25] A.R. Prabowo, D.M. Bae, J.M. Sohn, A.F. Zakki, B. Cao, Q. Wang, Analysis of structural damage on the truck ship under side collision scenario, *Alex. Eng., J.* 57 (3) (2018) 1761–1771, <https://doi.org/10.1016/j.aej.2017.05.002>.
- [26] L. Zhu, X. Wang, Ship collision model tests in a water tank, *Mar, Struct* 103 (2025), <https://doi.org/10.1016/j.marstruc.2025.103791> no. October 2024.
- [27] B. Liu, R. Villavicencio, P.T. Pedersen, C.Guedes Soares, Analysis of structural crashworthiness of double-hull ships in collision and grounding, *Mar., Struct.* 76 (2021) 102898, <https://doi.org/10.1016/j.marstruc.2020.102898> no. September 2020.
- [28] A. Presura, Collision analysis of the unconventional double hull of a ship, *World. J. Eng., Technol.* 05 (04) (2017) 601–612, <https://doi.org/10.4236/wjet.2017.54051>.
- [29] C. Jiang, L. Lin, N. Chen, A study on auxetic-inspired side structure for enhanced crashworthiness, *Mar., Struct.* 93 (2024) 103545, <https://doi.org/10.1016/j.marstruc.2023.103545> no. August 2023.
- [30] A.R. Mohammed, Z. Saleh, A.M. Aldabbagh, A. Al Hanbali, Multi-objective ergonomics design model optimization for micro electric cars via response surface methodology, *Discov. Appl. Sci.* 6 (10) (2024), <https://doi.org/10.1007/s42452-024-06219-z>.
- [31] I.J. Hogan Itam, D. Horsfall, I.T. Martin, E.C. Abam, Bibliometric approach to ‘applications of response surface methodology (RSM) publications’ and its review as the most widely used optimization tool, *Collnet J. Sci., Inf., Manag.* 18 (1) (2022), <https://doi.org/10.47974/cjsim-2022-0085>.
- [32] L. Adli, F. Cheggaga, N. Hannane, F. Ouzeri, Vertical wind speed extrapolation: modelling using a response surface methodology (RSM) based on unconventional designs, *Wind Eng* (2021), <https://doi.org/10.1177/0309524X211046380>.
- [33] W.D. Lestari, et al., Optimization of 3D printed parameters for socket prosthetic manufacturing using the taguchi method and response surface methodology, *Results Eng* 21 (2024), <https://doi.org/10.1016/j.rineng.2024.101847> no. October 2023.
- [34] J. Guerra, L. Velásquez, A. Rubio-Clemente, L. Jaramillo, E. Chica, Design and optimization of a siphon turbine using the response surface methodology, *Results Eng* 22 (2024), <https://doi.org/10.1016/j.rineng.2024.102241> no. May.
- [35] J. Lee, J.-K. Jung, D.-H. Lim, J. Lee, K.-D. Lee, A study on the synchronous reluctance motor design for high torque by using RSM, *IEEE Trans., Magn.* 54 (3) (2018) 1–5, <https://doi.org/10.1109/TMAG.2017.2749325>.
- [36] R. M. Lostado-Lorza, S. Ruiz-González, C. Sabando-Fraile, Corral-Bobadilla, Shielded metal arc welding (SMAW): determining the thermal fields with FEM and RSM, 2024 9th Int., Conf., Smart Sustain., Technol. (2024), <https://doi.org/10.23919/splitech61897.2024.10612553>.
- [37] M.F. Guo, H. Yang, L. Ma, 3D lightweight double arrow-head plate-lattice auxetic structures with enhanced stiffness and energy absorption performance, *Compos., Struct.* 290 (2022), <https://doi.org/10.1016/j.compstruct.2022.115484> no. December 2021.
- [38] G.S. Balan, S.A. Raj, Impact load bearing and energy absorption in sandwich polymer composites: projectile versus shock loadings, *Results. Eng* 25 (2025), <https://doi.org/10.1016/j.rineng.2024.103815> no. November 2024.
- [39] A. Al Ali, E. Arhore, H. Ghasemnejad, M. Yasaee, Experimental and numerical investigation of multi-layered honeycomb sandwich composites for impact mechanics applications, *Results. Eng* 21 (2024), <https://doi.org/10.1016/j.rineng.2024.101817> no. January.
- [40] M.B. Francisco, J.L.J. Pereira, G.A.V.B. Vasconcelos, S.S. da C. Jr, and G.F. Gomes, “Multiobjective design optimization of double arrowhead auxetic model using Lichtenberg algorithm based on metamodeling,” vol. 45, no. September, pp. 1199–1211, 2022, [doi:10.1016/j.istruc.2022.09.091](https://doi.org/10.1016/j.istruc.2022.09.091).
- [41] Q. Gao, C. Ge, W. Zhuang, L. Wang, Z. Ma, Crashworthiness analysis of double-arrowed auxetic structure under axial impact loading, *Mater., Des.* 161 (2019) 22–34, <https://doi.org/10.1016/j.matdes.2018.11.013>.
- [42] O. Khaghani, D. Mostofinejad, S.M. Abtahi, Auxetic structures in civil engineering applications : experimental (by 3D printing) and numerical investigation of mechanical behavior, *Results Mater* 21 (2024) 100528, <https://doi.org/10.1016/j.rinma.2024.100528> no. December 2023.
- [43] M. Hussain, S.M. Khan, M. Shafiq, N. Abbas, A review on PLA-based biodegradable materials for biomedical applications, *Giant* 18 (2024), <https://doi.org/10.1016/j.giant.2024.100261>.
- [44] P. Mondal, M.Avinash Mohan, R. Jayaganthan, Review of mechanical properties and impact response of PLA auxetic structures, *Mater. Today Proc* 87 (2023) 307–313, <https://doi.org/10.1016/j.matpr.2023.06.084>.
- [45] N. Maqsood, M. Rimašauskas, Characterization of carbon fiber reinforced PLA composites manufactured by fused deposition modeling, *Compos. C Open Access* 4 (2021), <https://doi.org/10.1016/j.jcomc.2021.100112> no. November 2020.
- [46] V.C. Gavalí, P.R. Kubade, H.B. Kulkarni, Property enhancement of carbon Fiber reinforced polymer composites prepared by fused deposition modeling, *Mater. Today Proc* 23 (2020) 221–229, <https://doi.org/10.1016/j.matpr.2020.02.020>.
- [47] M. Singh, et al., Targeted mechanical and energy absorption properties of 3D printed aluminium metamaterials, *Next Mater* 7 (2025) 100356, <https://doi.org/10.1016/j.nxmate.2024.100356> no. July 2024.
- [48] R. Johnston, Z. Kazanci, Analysis of additively manufactured (3D printed) dual-material auxetic structures under compression, *Addit, Manuf.* 38 (2021), <https://doi.org/10.1016/j.addma.2020.101783> no. September 2020.
- [49] F.A.F. Astuti, M.A. Choiron, A. Purnowidodo, Y.S. Irawan, Energy absorption and deformation pattern of honeycomb hybrid crash box under frontal load, *Proc, Transdiscipl., Symp., Eng., Technol.* (2024), <https://doi.org/10.1063/5.0201851>.
- [50] E. TUNCA, H. KAFALI, Compression and three-point bending analyzes of aerospace sandwich composites produced with polymeric core materials with ANSYS, *Eur., J. Sci., Technol.* 31 (31) (2021) 553–561, <https://doi.org/10.31590/ejosat.1012658>.
- [51] L. Wang, X. Jiang, M. Guo, X. Zhu, B. Yan, Characterisation of structural properties for AlSi10Mg alloys fabricated by selective laser melting, *Mater., Sci., Technol., (U. K.)* 33 (18) (2017) 2274–2282, <https://doi.org/10.1080/02670836.2017.1398513>.
- [52] F.A.F. Astuti, M.A. Choiron, A. Purnowidodo, Y.S. Irawan, Quasi-static test data set of honeycomb hybrid tubes by FDM 3D printing, *Data Br* 58 (2025) 111233, <https://doi.org/10.1016/j.dib.2024.111233>.
- [53] X. Fan, et al., Impact resistance performance and comparative analysis of CFRP and steel strand anchors, *Results Eng* 25 (2025), <https://doi.org/10.1016/j.rineng.2024.103511> no. November 2024.
- [54] H. Junaedi, T. Khan, W. Abd-Elaziem, T.A. Sebaey, K. Akkad, Optimization of circular hole patterns on crashworthiness of polyurethane foam-filled CFRP rectangular tubes, *Results Eng* 26 (2025), <https://doi.org/10.1016/j.rineng.2025.104805> no. March.
- [55] A.A. Garcia-Granada, High-compression crash simulations and tests of PLA cubes fabricated using additive manufacturing FDM with a scaling strategy, *Computation* 12 (3) (2024), <https://doi.org/10.3390/computation12030040>.
- [56] R.F. Faidallah, A.M. Abd-El Nabi, M.M. Hanon, Z. Szakál, I. Oldal, Compressive and bending properties of 3D-printed wood/PLA composites with re-entrant honeycomb core, *Results Eng* 24 (2024), <https://doi.org/10.1016/j.rineng.2024.103023> no. September.
- [57] X. Cai, et al., Mechanical behavior, damage mode and mechanism of AlSi10Mg porous structure manufactured by selective laser melting, *J. Alloys Compd* 897 (2022), <https://doi.org/10.1016/j.jallcom.2021.162933>.
- [58] A. Baroutaji, A. Arjunan, G. Singh, J. Robinson, Crushing and energy absorption properties of additively manufactured concave thin-walled tubes, *Results Eng* 14 (2022), <https://doi.org/10.1016/j.rineng.2022.100424> no. September 2021.
- [59] J. Liu, W. Yang, J. Liu, J. Liu, W. Huang, Ballistic impact analyses of foam-filled double-arrow auxetic structure, *Thin-Walled Struct.* 182 (2023), <https://doi.org/10.1016/j.tws.2022.110173> no. June 2022.
- [60] R. Cavenagh, et al., Influence of the cover plate thickness on the ballistic penetration of re-entrant auxetic structures, *Int. J. Impact Eng.* 186 (2024) 104890, <https://doi.org/10.1016/j.ijimpeng.2024.104890> no. January.
- [61] M. Chen, S. Fang, G. Wang, Y. Xuan, D. Gao, M. Zhang, Compressive and flexural behaviour of engineered cementitious composites based auxetic structures: an experimental and numerical study, *J. Build. Eng.* 86 (2024) 108999, <https://doi.org/10.1016/j.jobbe.2024.108999> no. January.
- [62] C. Li, Q. Zhou, H. Li, and R. Wang, “Dynamic crushing responses of enhanced auxetic re-entrant honeycomb based on additive manufacturing,” vol. 69, no. September, 2024.
- [63] M. Rehan, T. He, D. Tahir, A.K. Khalil, W.S. Yip, S.S. To, 3D Finite element modeling and experimental investigation of micro milling of laser powder bed

- fusion fabricated Ti6Al4V, Results Eng. 26 (2025), <https://doi.org/10.1016/j.rineng.2025.104635> no. November 2024.
- [64] M.T. Nasri, F. Abbassi, T. Hamdeni, M. Demiral, M.A. Rezgui, M. Ayadi, Novel approach for sheet metal constitutive parameters identification based on shape index and multiple regression, Results Eng 22 (2024), <https://doi.org/10.1016/j.rineng.2024.102267> no. May.
- [65] R. Setiawan, A.I. Komara, B. Budiwardoro, Numerical and experimental investigation of aluminum double arrowhead cellular structure manufactured using sheet metal forming, Results Eng 26 (2025), <https://doi.org/10.1016/j.rineng.2025.104667> no. March.
- [66] M. Jimenez-martinez, J. Varela-soriano, J. Martinez-trinidad, M. Alfaro-ponce, M. C.-Gonzalez, Enhancing energy absorption in printed PLA components through post-processing and topological optimization, Results Eng 26 (2025) no. April.
- [67] G. Luo, J. Liu, L. Li, P. Xue, J. Zhao, Y. Chen, Crashworthiness analysis and multi-objective optimization design for foam-filled spiral tube, Int., J. Mech., Sci. 282 (2024) 109588, <https://doi.org/10.1016/j.ijmecsci.2024.109588> no. June.
- [68] C.W. Isaac, C. Ezekwem, A review of the crashworthiness performance of energy absorbing composite structure within the context of materials, manufacturing and maintenance for sustainability, Compos. Struct. 257 (2021) 113081, <https://doi.org/10.1016/j.compstruct.2020.113081> no. May 2020.
- [69] S.L.C. Ferreira, et al., Box-Behnken design: an alternative for the optimization of analytical methods, Anal. Chim. Acta 597 (2) (2007), <https://doi.org/10.1016/j.aca.2007.07.011>.
- [70] A. Mengistu, G. Andualem, M. Abewaa, D. Birhane, Keratin extraction optimization from poultry feather using response surface- Box-Behnken experimental design method, Results Eng 22 (2024), <https://doi.org/10.1016/j.rineng.2024.102360> no. April.
- [71] S. Bhardwaj, D.K. Mishra, Effect of different strut thickness on the mechanical behaviour of 3D printed auxetic structures, Mater, Today Proc. (2024), <https://doi.org/10.1016/j.matpr.2024.04.096> no. April.
- [72] A. sadat Mirhakimi, D. Dubey, M.A. Elbestawi, Laser powder bed fusion of bio-inspired metamaterials for energy absorption applications: a review, J. Mater, Res. Technol. 31 (2024) 2126–2155, <https://doi.org/10.1016/j.jmrt.2024.06.234>, no. July.
- [73] Jianjun Zhang, G. Lu, Z. You, Large deformation and energy absorption of additively manufactured auxetic materials and structures: a review, Compos. B Eng. 201 (2020) 1–90, <https://doi.org/10.1016/j.compositesb.2020.108340>.
- [74] Q. Gao, X. Zhao, C.Z. Wang, L.M. Wang, Z.D. Ma, Crashworthiness analysis of a cylindrical auxetic structure under axial impact loading, Sci., China Technol, Sci 63 (1) (2020) 140–154, <https://doi.org/10.1007/s11431-018-9467-6>.
- [75] S. Bronder, et al., Hybrid auxetic structures: structural optimization and mechanical characterization, Adv., Eng., Mater. 23 (5) (2021), <https://doi.org/10.1002/adem.202001393>.
- [76] S.S. Vyavahare, S. Kumar, Numerical and experimental investigation of FDM fabricated re-entrant auxetic structures of ABS and PLA materials under compressive loading, Rapid Prototyp., J. (2021), <https://doi.org/10.1108/RPJ-10-2019-0271>.
- [77] A.C. Kaya, A. Korucu, and M. Boğuçlu, “Influence of additional strut elements in 3D re-entrant auxetic unit.” 2024.
- [78] J. Kim, D.G. Kim, K.H. Ryu, Enhancing response surface methodology through coefficient clipping based on prior knowledge, Processes 11 (12) (2023) 1–14, <https://doi.org/10.3390/pr11123392>.
- [79] M. Lotfi, A. Masoumi, Multi-objective crashworthiness optimization for a newly developed 3D re-entrant auxetic structure using response surface method and MOPSO algorithm, Proc. Inst., Mech., Eng., D J. Automob., Eng. (2024) 1–11, <https://doi.org/10.1177/09544070231221015>.
- [80] M. Najafi, H. Ahmadi, G. Liaghat, Experimental investigation on energy absorption of auxetic structures, Mater., Today Proc. 34 (2019) 350–355, <https://doi.org/10.1016/j.matpr.2020.06.075>.
- [81] T. Shepherd, K. Winwood, P. Venkatraman, A. Alderson, T. Allen, Validation of a finite element modeling process for auxetic structures under impact, Phys. Status Solidi Basic Res. 257 (10) (2020), <https://doi.org/10.1002/pssb.201900197>.
- [82] Q. Wang, Y. Zeng, H. Nie, B. Jia, F. Gao, Energy absorption characteristics of carbon fiber reinforced plastic/aluminum hybrid materials double arrow-head auxetic structure, J. Sandw, Struct, Mater 26 (4) (2024), <https://doi.org/10.1177/10996362241226980>.
- [83] W. Jiang, W. Ao, Y. Liu, J. Liu, J. Huang, Dynamic responses and failure mechanism of composite double-arrow auxetic structure under impact loading, Mech., Adv., Mater., Struct. 30 (13) (2022), <https://doi.org/10.1080/15376494.2022.2059600>.
- [84] M. Safikhani Nasim, A. Yaghootian, R. Mosalmani, Energy absorption of the additively manufactured novel re-entrant auxetic structure in comparison with honeycomb structure: experimental and numerical analysis, J. Braz. Soc., Mech., Sci., Eng. 45 (5) (2023) 1–24, <https://doi.org/10.1007/s40430-023-04178-x>.
- [85] J. Dong, G. Ye, Y. Wang, F. Jin, H. Fan, Design, manufacture and crushing behaviors of buckling-inspired auxetic meta-lattice structures, Int., J. Smart Nano Mater 12 (4) (2021) 491–510, <https://doi.org/10.1080/19475411.2021.1966855>.
- [86] M.E. Yazıcı, B. Kanber, S. Dağlı, Investigation of bending behavior of two and three-dimensional honeycomb and auxetic sandwich beams, J. Sandw., Struct., Mater. 26 (6) (2024) 945–968, <https://doi.org/10.1177/10996362241257871>.
- [87] D. Mocerino, M.R. Ricciardi, V. Antonucci, I. Papa, Fused deposition modelling of polymeric auxetic structures: a review, Polym. (Basel) 15 (4) (2023) 1–17, <https://doi.org/10.3390/polym15041008>.